

Reviewer Pack — Minimal Modular Forms Order Correlation with Boolean Circuit...

Entry 5d0e1577b38c · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Modular Forms Order Correlation with Boolean Circuit Weights

Entry ID: 5d0e1577b38c **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 17:15:45 UTC

1. Conjecture

Field A (mathematical branch): Modular Form Theory **Field B** (complexity object): Boolean Circuit Complexity

Statement:

For every boolean circuit C with size s , the minimal order of an eigenform associated with C 's associated matrix A is linearly correlated with its weight $w(C)$, such that $\text{MinimalOrder}(A) = \Theta(w(C))$.

Rationale (proposer's reasoning):

Modular forms encode arithmetic structure and their orders are closely related to number-theoretic properties. The weights of boolean circuits reflect their computational complexity, suggesting a potential link between these two seemingly unrelated fields.

Taxonomy category: MODULAR_FORMS_TO_CIRCUIT_WEIGHT_CORRELATION (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): cd696730b48a3c5f

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between $\text{MinimalOrder}(A)$ and $w(C)$ for all circuits with size $s \leq 40$ is greater than or equal to 0.8, and the mean absolute deviation of $\text{MinimalOrder}(A)$ from $\Theta(w(C))$ across all circuits is less than or equal to 3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 2 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Modular Forms AND Boolean Circuit Complexity - Eigenform AND weight correlation IN Boolean Circuit Theory AND Modular Form Theory - Boolean Circuit Size AND Eigenform Order θ -linear relation

Top relevant hits considered: - [s2:10.20295/2412-9186-2025-11-02-178-193] Research into the Properties of Modular Weight-Based Sum Codes in Concurrent Error-Detection Circuits Based on Boolean S - [s2:10.1002/cta.4265] Lightweight High-Throughput TRNG Based on Single-Node Boolean Chaotic Structure

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(A):
    n = len(A)
    for i in range(n):
        # Find pivot row
        max_row = i
        for r in range(i+1, n):
            if abs(A[r][i]) > abs(A[max_row][i]):
                max_row = r
        A[i], A[max_row] = A[max_row], A[i]

        # Eliminate non-zero entries below pivot
        factor = A[i][i]
        for j in range(i, n):
            A[i][j] /= factor

        for k in range(i+1, n):
            factor = A[k][i]
            for j in range(i, n):
                A[k][j] -= factor * A[i][j]

def min_eigenform_order(A):
    gaussian_elimination(A)
    order = 0
    for row in A:
        if any(x != Fraction(0) for x in row):
```

```

        order += 1
    return order

def boolean_circuit_weight(C):
    # Placeholder function to compute the weight of a boolean circuit
    # This is a dummy implementation and should be replaced with actual logic
    return len(C)

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 40
    instances_tested = 30
    metric_value = 0.0
    counterexample = ""
    conjecture_holds = False

    for _ in range(instances_tested):
        # Generate a random boolean circuit with size  $s \leq 40$ 
        C = [random.choice([0, 1]) for _ in range(n)]

        # Compute the associated matrix A for each circuit C
        A = [[C[i] * C[j] for j in range(n)] for i in range(n)]

        # Determine the minimal order of an eigenform associated with A, MinimalOrder(A)
        order = min_eigenform_order(A)

        # Calculate the weight  $w(C)$  of each circuit C
        weight = boolean_circuit_weight(C)

        # Compute the correlation between MinimalOrder(A) and  $w(C)$  using Pearson correlation coefficient
        metric_value += order * weight

    mean_metric_value = metric_value / instances_tested
    return {
        "metric_name": "Pearson Correlation Coefficient",
        "metric_value": mean_metric_value,
        "instances_tested": instances_tested,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

```

```

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9a3b9d1.py", line 92, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9a3b9d1.py", line 68, in run_trial
    order = min_eigenform_order(A)
            ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9a3b9d1.py", line 39, in min_eigenform_order
    gaussian_elimination(A)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b9a3b9d1.py", line 31, in gaussian_elimination
    A[i][j] /= factor
ZeroDivisionError: float division by zero

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed due to a ZeroDivisionError, preventing the calculation of the Pearson correlation coefficient and mean absolute deviation required to evaluate the conjecture. | next: Investigate the cause of the ZeroDivisionError in the gaussian_elimination function and ensure that division by zero does not occur. Once this is resolved, rerun the test with a valid dataset to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13232
2	preregistration	ollama_remote	glm4:latest	0	0	10022
3	novelty	ollama_remote	glm4:latest	0	0	9798

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
4	novelty	ollama_remote	glm4:latest	0	0	20716
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20626
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21195
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16222
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17401
9	judge	ollama_remote	glm4:latest	0	0	11311

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 140522 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/5d0e1577b38c.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/5d0e1577b38c.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/5d0e1577b38c.tar.gz` (if generated)